

Data Quality Diagnostic Report

Source: sample_sales.xlsx
May 15, 2026



4 findings across 15 columns and 500 rows

Executive Summary

No critical issues were found, but 1 warning and 3 informational observations were detected. Address the warnings before using this data in production.

Detailed Findings

Warning (1)

Warning	revenue	Cardinality Anomaly
Assumption	Column 'revenue' appears to be a unique identifier (ID column).	
Reality	However, 496 of 500 values are unique (uniqueness ratio: 99.2%), meaning some IDs appear more than once. Duplicate values include: '117.71', '210.24', '263.77', '9.52'.	
Impact	Duplicate IDs in 'revenue' cause row-level joins to fan out, producing unexpected extra rows in merged datasets. Aggregations that assume one row per ID will double-count affected records.	
Recommended Fix	Investigate the duplicate values in 'revenue' to determine whether they are true duplicates (same record entered twice) or legitimate repeats (one-to-many relationship). De-duplicate or re-model accordingly.	
Prevention Rule	If 'revenue' is meant to be a primary key, enforce a uniqueness constraint at the database or validation layer.	

Info (3)

Info	constant_col	Cardinality Anomaly
Assumption	Column 'constant_col' is expected to carry meaningful, varying data.	
Reality	However, only 1 distinct value was found across 500 rows (uniqueness ratio: 0.2%). Most common: 'FIXED' (500 times).	
Impact	A near-constant column like 'constant_col' adds no analytical value. Including it in models or reports may mislead readers into thinking the field varies when it does not.	
Recommended Fix	Verify whether 'constant_col' should actually vary. If not, document it as a constant and consider removing it from analysis. If it should vary, investigate why the data is uniform.	
Prevention Rule	If 'constant_col' is supposed to carry diverse values, add a data-quality check that flags columns with fewer than 1% unique values.	

Info	is_renewal	Cardinality Anomaly
Assumption	Column 'is_renewal' is expected to carry meaningful, varying data.	
Reality	However, only 2 distinct values were found across 500 rows (uniqueness ratio: 0.4%). Most common: 'True' (266 times), 'False' (234 times).	
Impact	A near-constant column like 'is_renewal' adds no analytical value. Including it in models or reports may mislead readers into thinking the field varies when it does not.	
Recommended Fix	Verify whether 'is_renewal' should actually vary. If not, document it as a constant and consider removing it from analysis. If it should vary, investigate why the data is uniform.	
Prevention Rule	If 'is_renewal' is supposed to carry diverse values, add a data-quality check that flags columns with fewer than 1% unique values.	

Info	order_status	Cardinality Anomaly
Assumption	Column 'order_status' is expected to carry meaningful, varying data.	
Reality	However, only 4 distinct values were found across 500 rows (uniqueness ratio: 0.8%). Most common: 'Pending' (136 times), 'Closed' (131 times), 'Active' (127 times).	
Impact	A near-constant column like 'order_status' adds no analytical value. Including it in models or reports may mislead readers into thinking the field varies when it does not.	
Recommended Fix	Verify whether 'order_status' should actually vary. If not, document it as a constant and consider removing it from analysis. If it should vary, investigate why the data is uniform.	
Prevention Rule	If 'order_status' is supposed to carry diverse values, add a data-quality check that flags columns with fewer than 1% unique values.	

Field Inventory

Summary of all fields with detected issues.

Field Name	Issue Types	Highest Severity	Finding Count
constant_col	Cardinality Anomaly	Info	1
is_renewal	Cardinality Anomaly	Info	1
order_status	Cardinality Anomaly	Info	1
revenue	Cardinality Anomaly	Warning	1